

Multi-Region Disaster Recovery Plan

Project: Multi-Region Disaster Recovery Plan

Services: EC2, RDS, Route 53, CloudFormation

Description: Set up a disaster recovery architecture where applications and database services are deployed across two AWS regions. Configure automated failover using Route 53 health checks to redirect traffic from the primary region to the disaster recovery region during failures.

Skills: High availability, failover planning, disaster recovery, Route 53 routing, cross-region architecture, cloud automation.

Definition: The Multi-Region Disaster Recovery Plan is designed to provide high availability and business continuity for cloud-based applications hosted on AWS. The project uses Amazon EC2 instances and Amazon RDS databases deployed across two AWS regions: Mumbai as the primary region and Singapore as the disaster recovery region. Amazon Route 53 health checks continuously monitor the primary server and automatically redirect traffic to the disaster recovery server if a failure occurs. AWS CloudFormation is used to automate infrastructure deployment and management.

Scope of the Project: The scope of this project is to design and implement a disaster recovery architecture that ensures application availability even during regional outages or server failures. The project demonstrates multi-region deployment, automated failover mechanisms, and infrastructure automation using AWS services.

This project covers:

- Multi-Region Deployment: Deploying application servers and databases in Mumbai and Singapore regions.
- Disaster Recovery Architecture: Configuring a secondary recovery environment that becomes active during primary region failures.
- Automated Failover: Using Route 53 health checks and failover routing policies to automatically redirect traffic.
- Infrastructure Automation: Managing cloud resources using AWS CloudFormation templates.
- High Availability Planning: Ensuring continuous access to applications with minimal downtime.

Methodology:

1.Route 53 Failover Routing – Best Method

- Uses all required AWS services: EC2, RDS, Route 53, and CloudFormation.
- Provides automated DNS failover between primary and disaster recovery regions.
- Health checks continuously monitor the availability of the primary server.
- Automatically redirects users to the DR server during outages.
- Production-ready disaster recovery approach suitable for enterprise cloud environments.

- Scalable and reliable solution for high availability applications.

2.Manual Disaster Recovery Method – Alternative Method

- Uses EC2 instances and manual failover procedures between regions.
- Administrators manually update DNS or access the DR server during failures.
- Simpler to implement but requires human intervention.
- Lower automation compared to Route 53 failover routing.
- Suitable for small-scale testing or learning environments.

AWS Services used in the Project:

1.Amazon EC2 (Elastic Compute Cloud)

Definition: Amazon EC2 is a cloud computing service that provides scalable virtual servers for hosting applications and services.

Purpose: Hosts the application servers in both primary and disaster recovery regions.

Role: Runs Apache web servers in Mumbai and Singapore regions for application availability.

2.Amazon RDS (Relational Database Service)

Definition: Amazon RDS is a managed relational database service that simplifies database setup, management, and scaling.

Purpose: Provides managed MySQL database services for application data storage.

Role: Maintains application databases in primary and disaster recovery regions.

3.Amazon Route 53

Definition: Amazon Route 53 is a scalable Domain Name System (DNS) web service that provides routing policies, domain registration, and health monitoring.

Purpose: Monitors application availability and manages failover routing.

Role: Performs automated failover between Mumbai and Singapore servers using health checks.

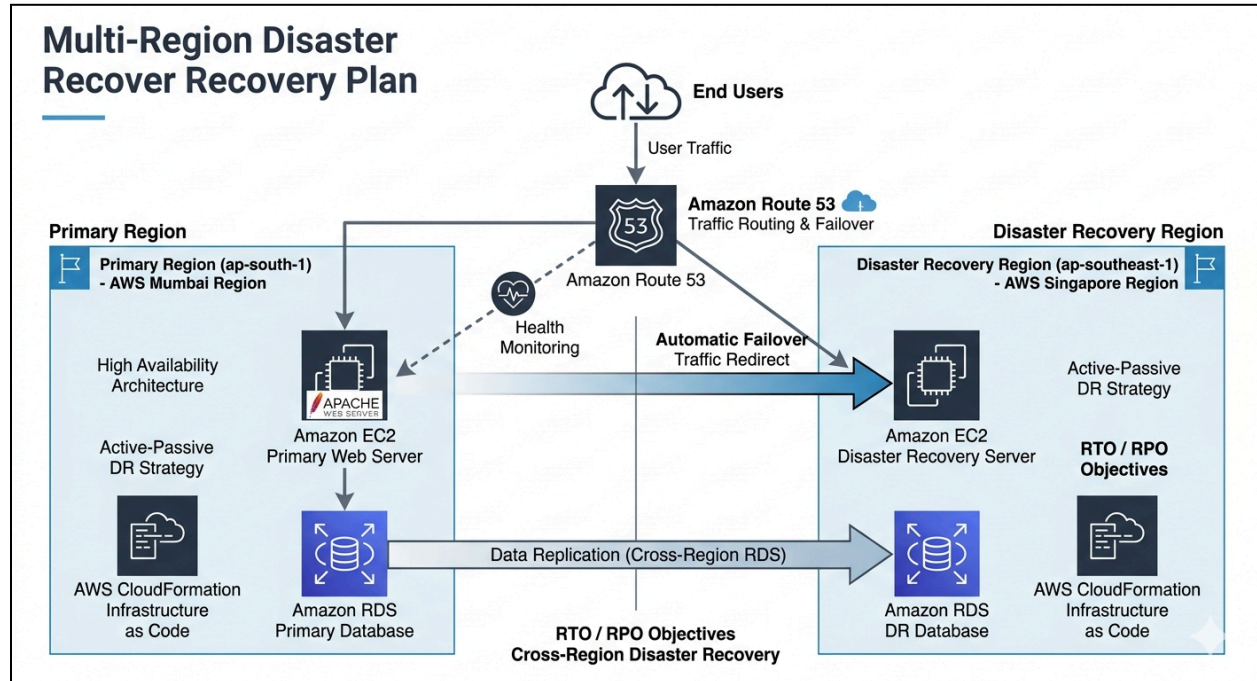
4.AWS CloudFormation

Definition: AWS CloudFormation is an Infrastructure as Code (IaC) service that automates cloud resource deployment using templates.

Purpose: Automates infrastructure provisioning and resource management.

Role: Creates and manages EC2 instances and related AWS infrastructure resources.

Architecture Diagram:

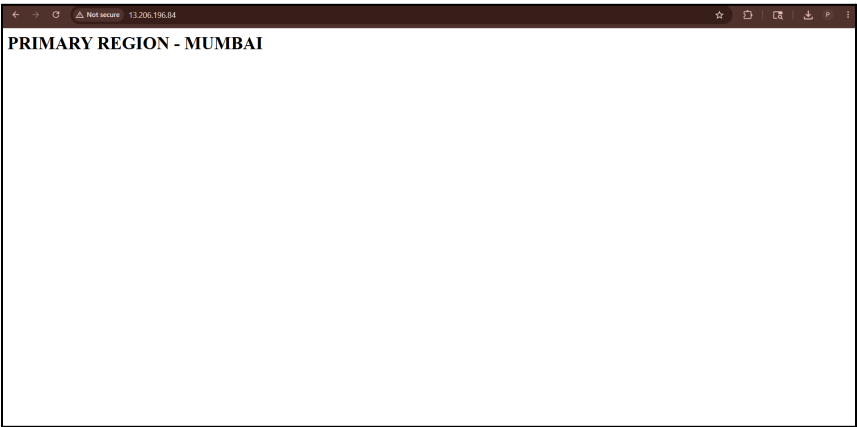
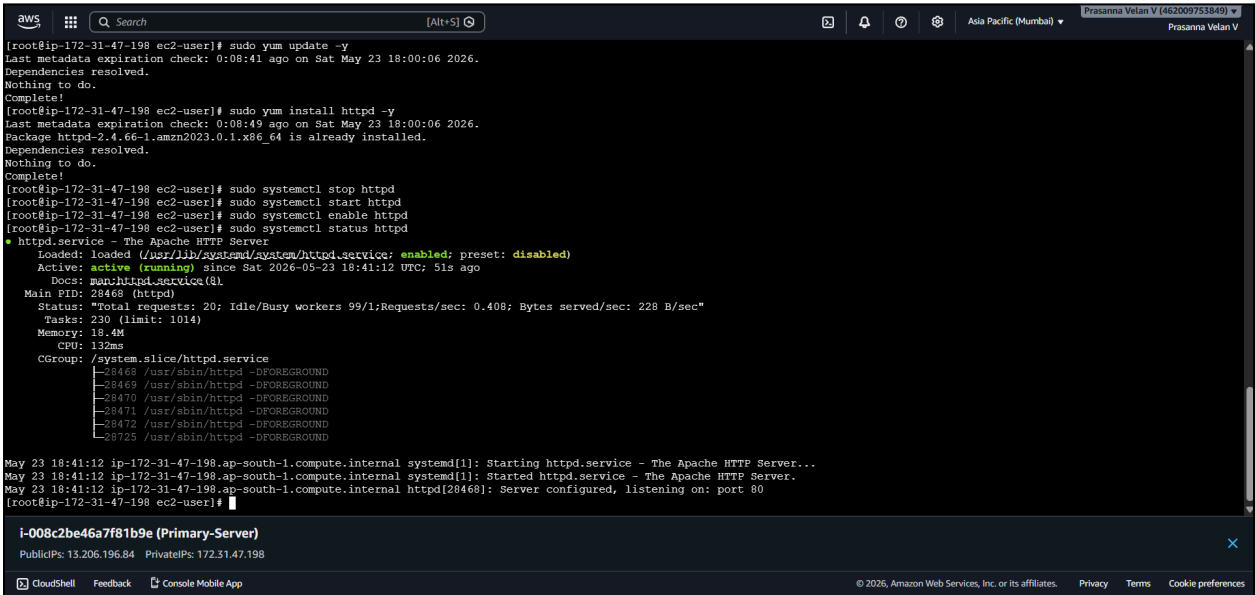
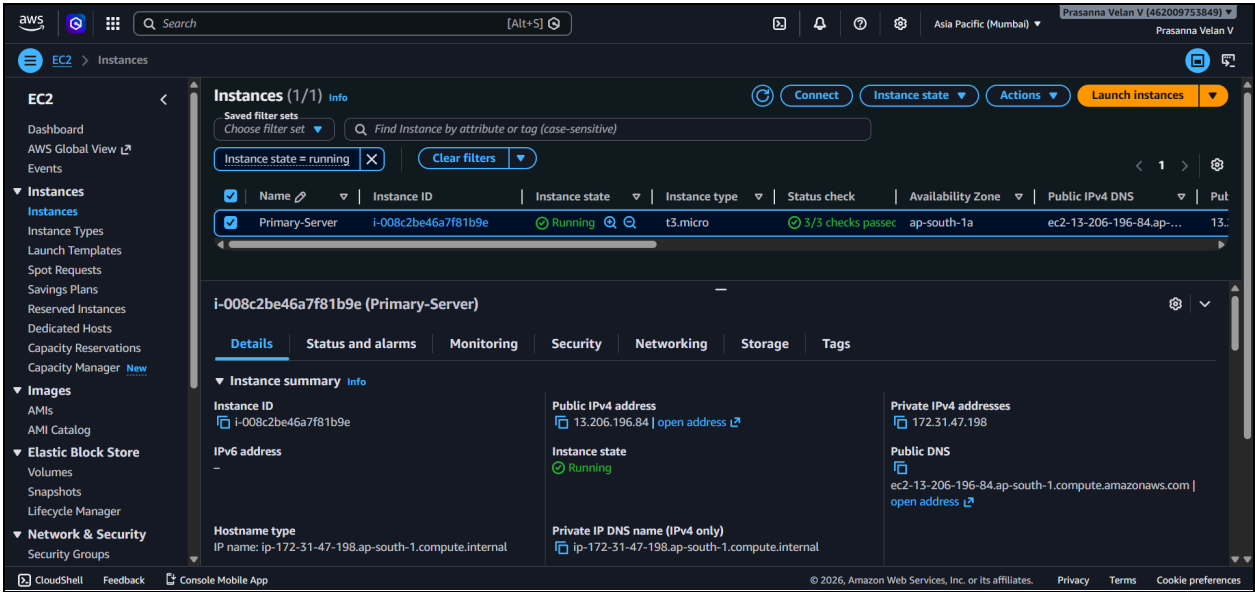


Implementation Steps to Do the Project:

Step 1: Launch Primary EC2 Instance in Mumbai Region

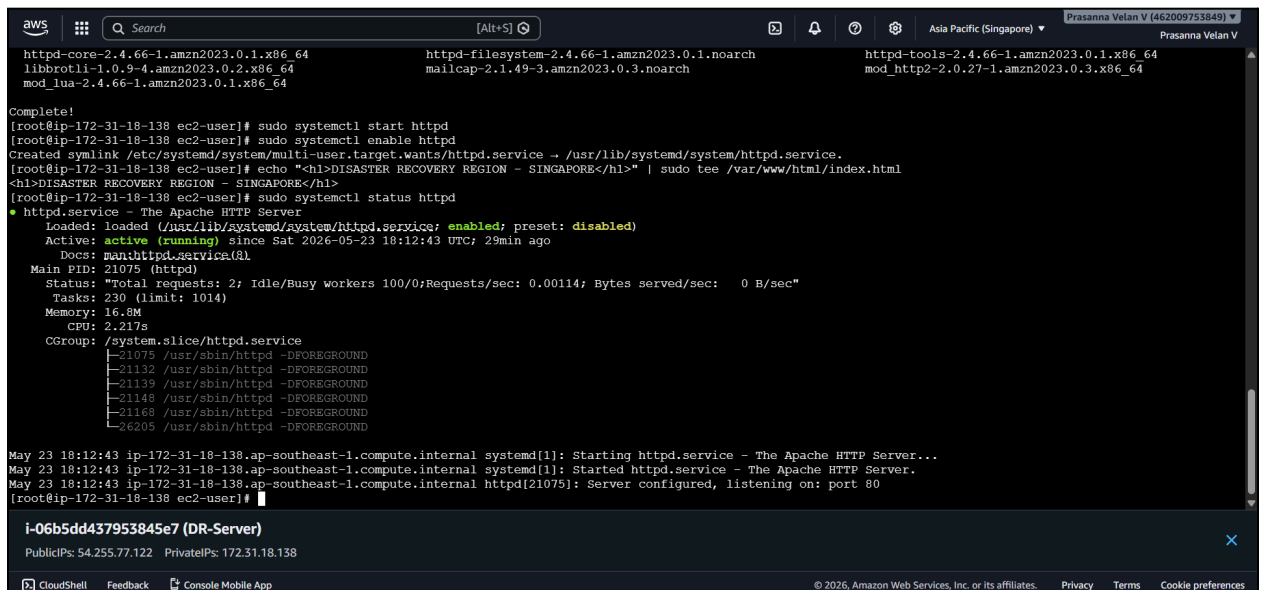
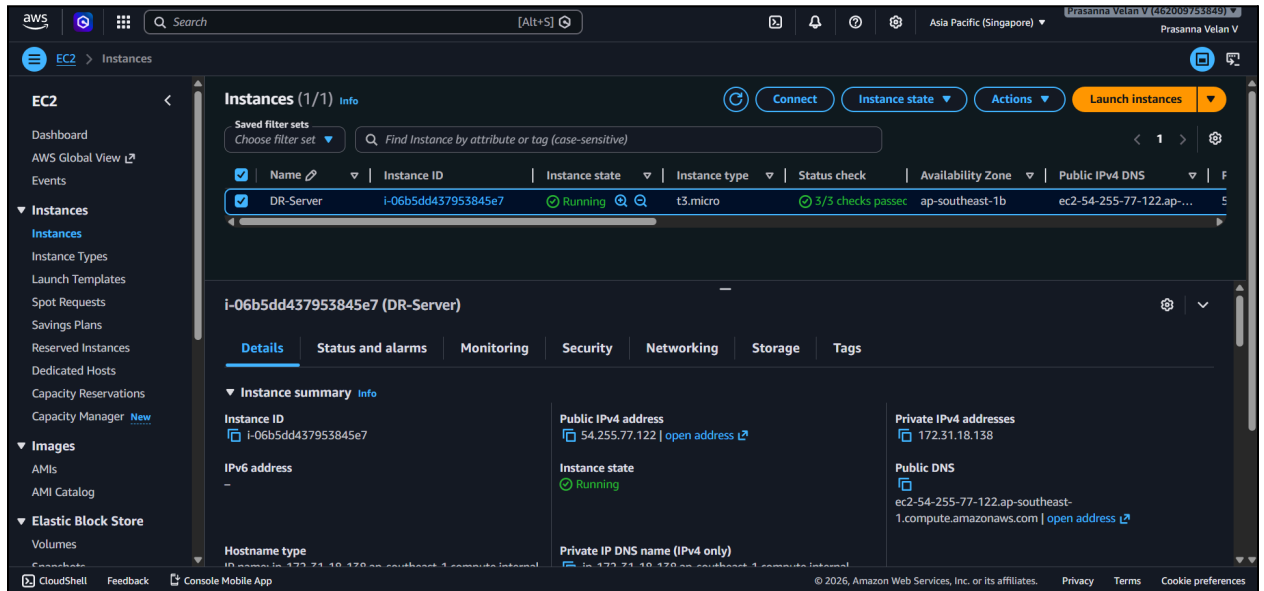
- Go to AWS Console → EC2 → Launch Instance.
- Select Mumbai region (ap-south-1).
- Choose Amazon Linux AMI.
- Configure instance type as t2.micro.
- Configure Security Group to allow SSH and HTTP access.
- Launch the EC2 instance.
- Install Apache web server and create a test webpage displaying “PRIMARY REGION - MUMBAI”.

Multi-Region Disaster Recovery Plan



Step 2: Launch Disaster Recovery EC2 Instance in Singapore Region

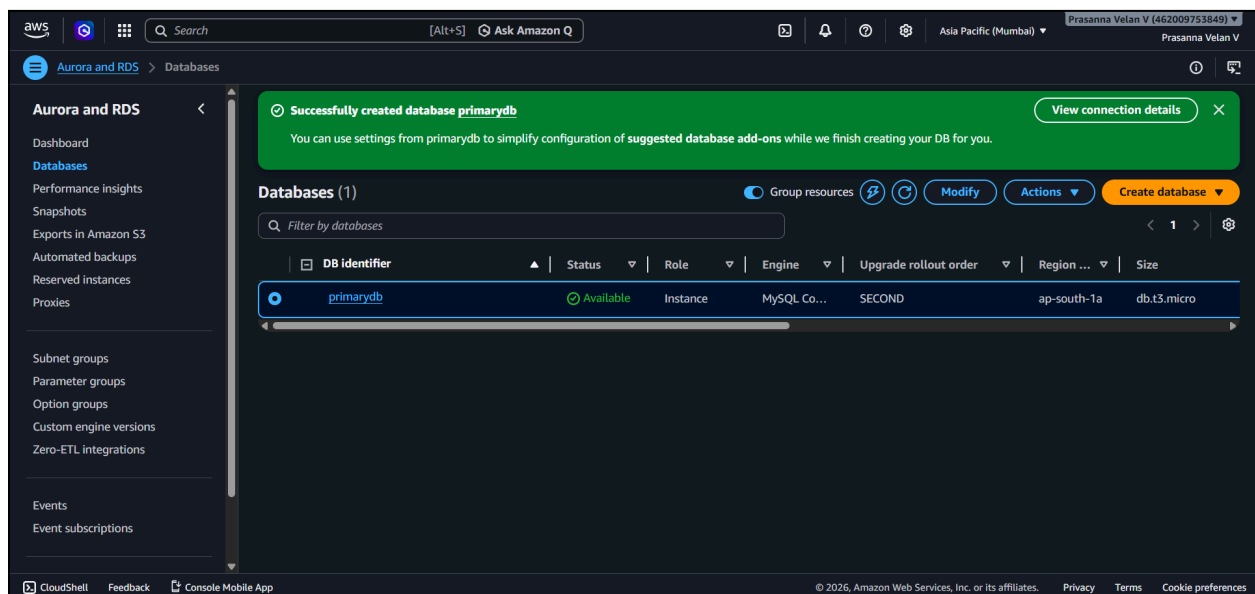
- Change AWS region to Singapore (ap-southeast-1).
- Launch another EC2 instance.
- Configure instance type as t2.micro.
- Configure Security Group to allow SSH and HTTP access.
- Install Apache web server and create a test webpage displaying “DISASTER RECOVERY REGION - SINGAPORE”.





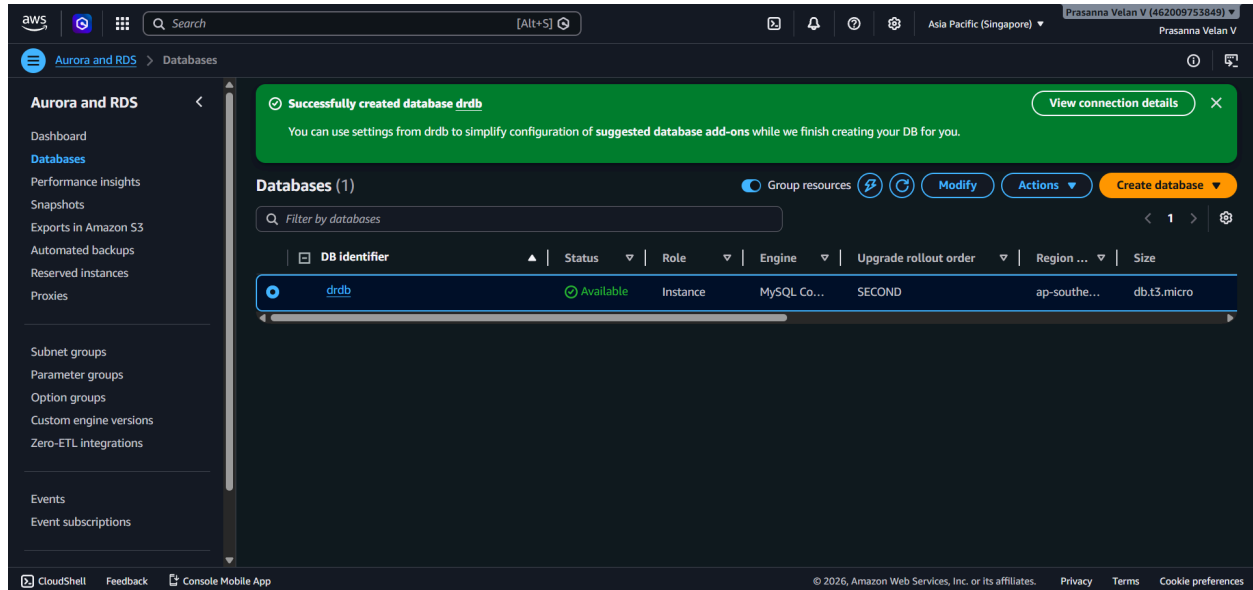
Step 3: Create RDS Database in Mumbai Region

- Go to AWS Console → RDS → Create Database.
- Select MySQL engine.
- Choose Free Tier template.
- Configure database instance as db.t3.micro.
- Create the primary database instance.



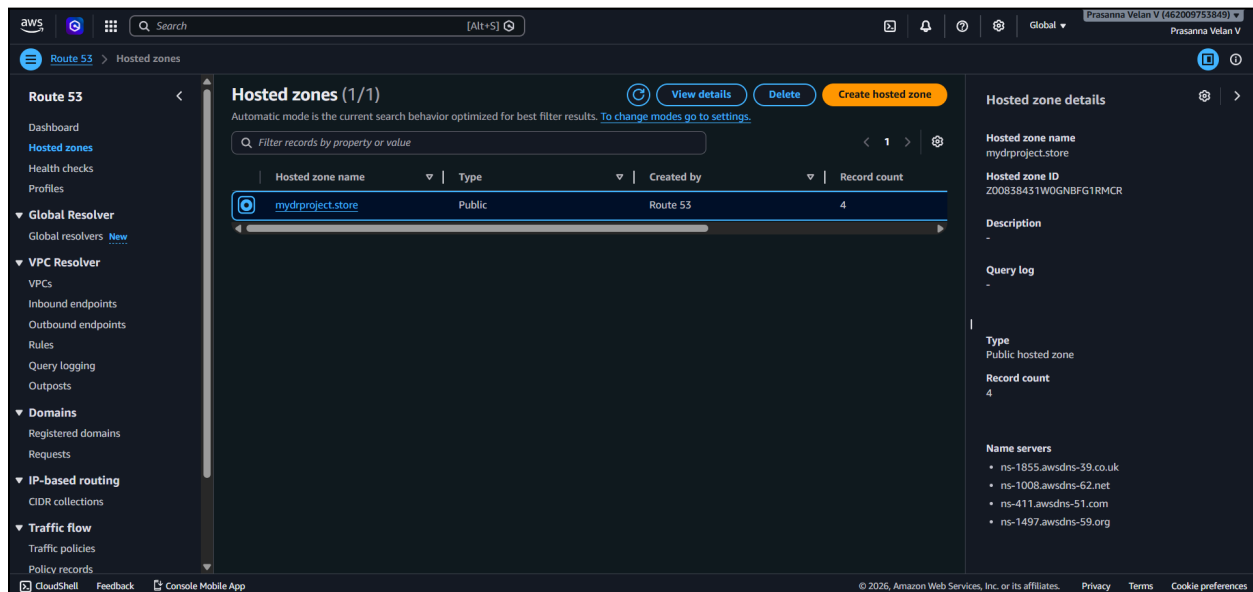
Step 4: Create RDS Database in Singapore Region

- Change AWS region to Singapore.
- Create another MySQL RDS database instance.
- Configure it as the disaster recovery database.



Step 5: Create Route 53 Hosted Zone

- Go to Route 53 → Hosted Zones.
- Create a new public hosted zone.
- Configure DNS management settings.



Step 6: Configure Route 53 Failover Records

- Create Primary failover A record pointing to Mumbai EC2 public IP.
- Attach Route 53 health check to the primary record.
- Create Secondary failover A record pointing to Singapore EC2 public IP.
- Configure failover routing policy.

Records (4) | Accelerated recovery | DNSSEC signing | Tags (0)

Records (4) [Info](#) [Refresh](#) [Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value [Type](#) [Routing p...](#) [Alias](#) < 1 > [Settings](#)

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...
☐	mydrproject.st...	A	Failover	Primary	No	13.206.196.84	300
☐	mydrproject.st...	A	Failover	Secondary	No	54.255.77.122	300
☐	mydrproject.st...	NS	Simple	-	No	ns-1855.awsdns-39.co.uk. ns-1008.awsdns-62.net. ns-411.awsdns-51.com. ns-1497.awsdns-59.org.	172800
☐	mydrproject.st...	SOA	Simple	-	No	ns-1855.awsdns-39.co.uk. a...	900

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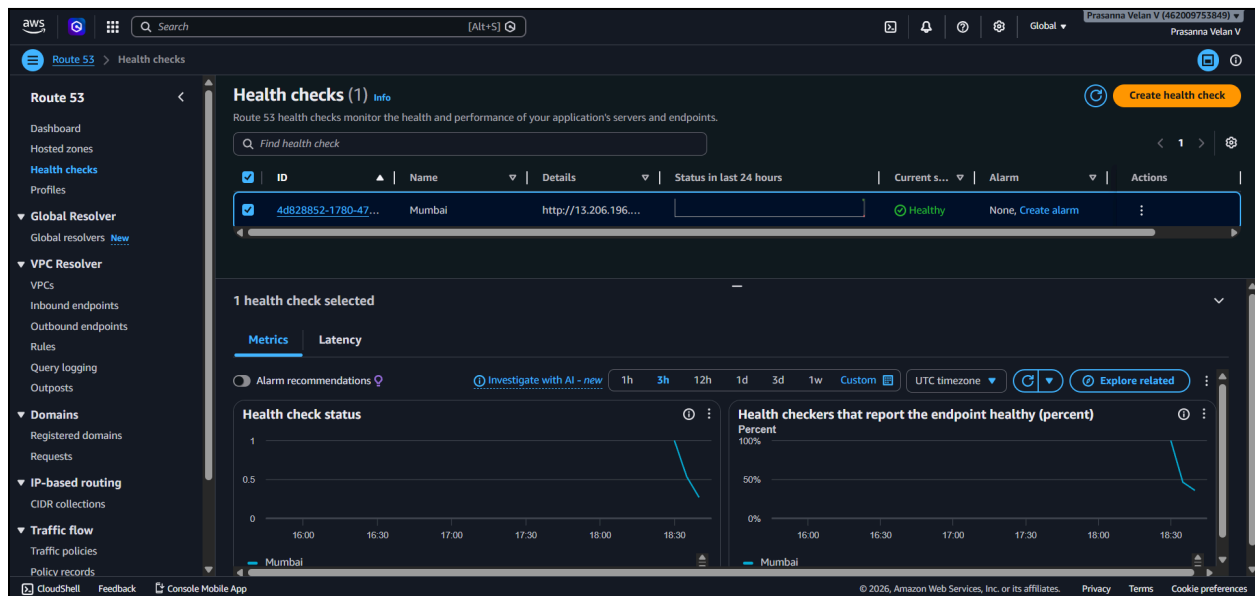
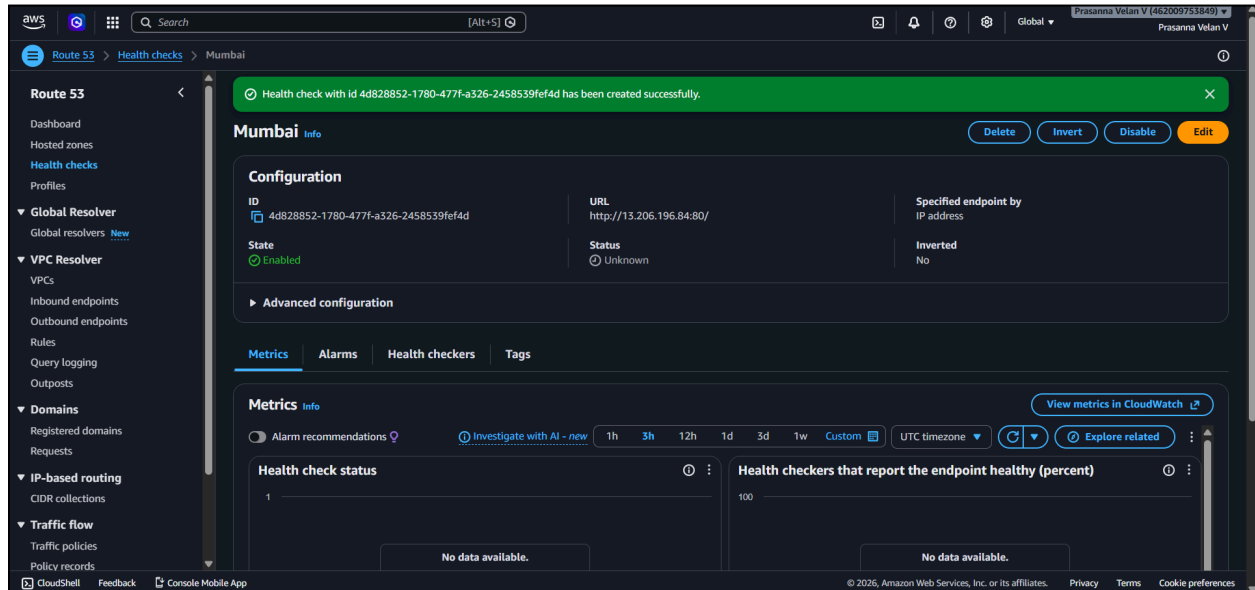
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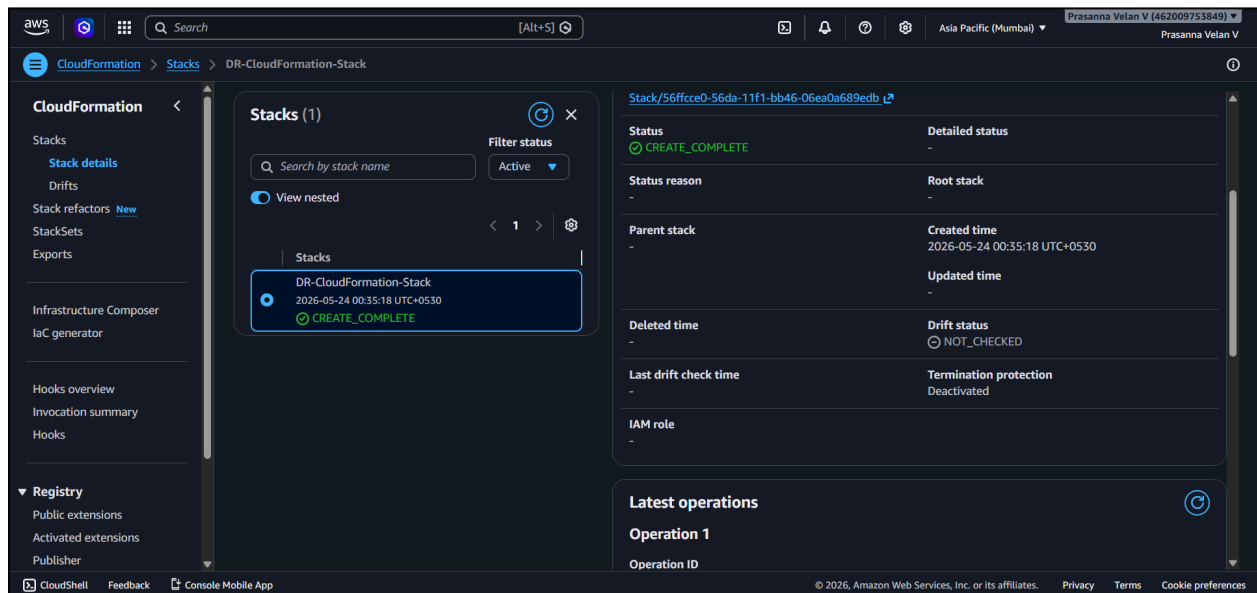
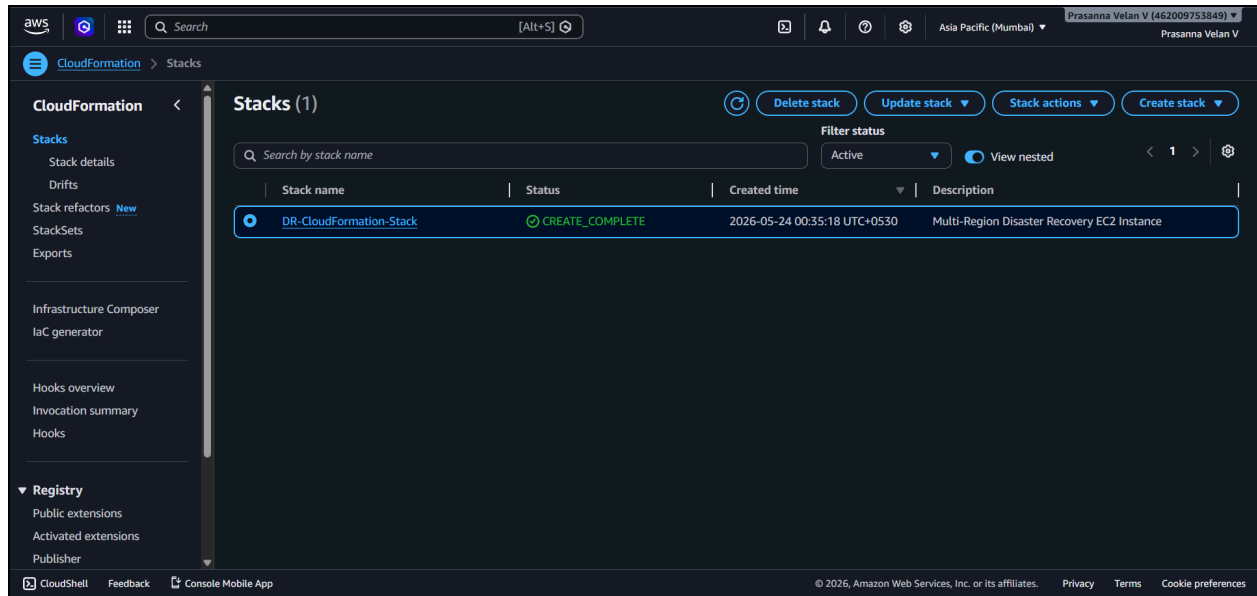
Step 7: Configure Route 53 Health Check

- Create a health check for the Mumbai EC2 instance.
- Configure protocol as HTTP.
- Set monitoring path as “/”.
- Verify health status changes between healthy and unhealthy conditions.



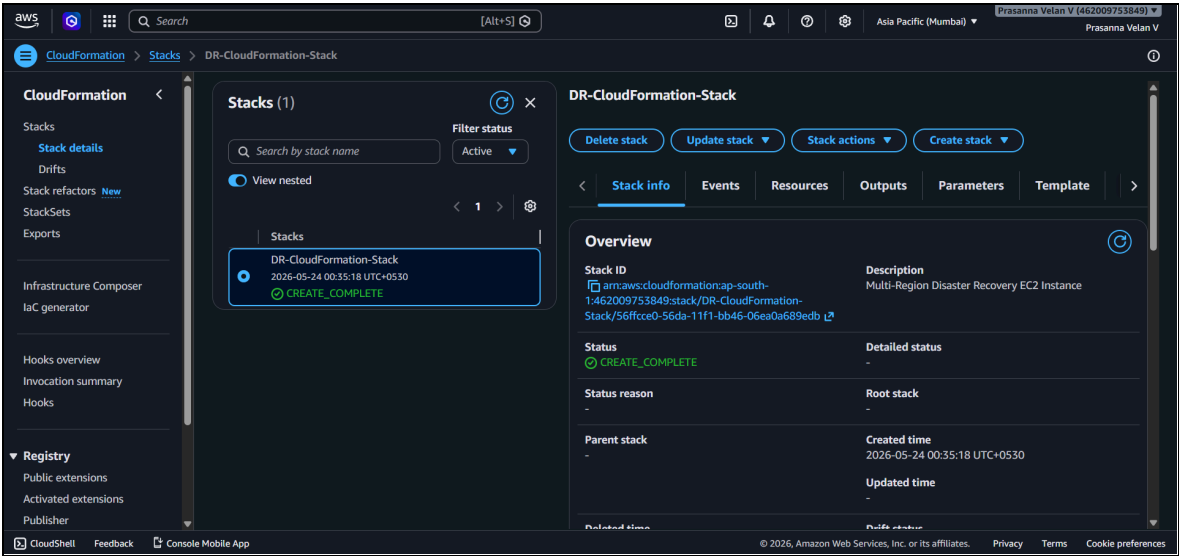
Step 8: Create CloudFormation Template

- Create a YAML template for EC2 resource deployment.
- Define EC2 instance properties in the template.
- Save the template file.



Step 9: Deploy CloudFormation Stack

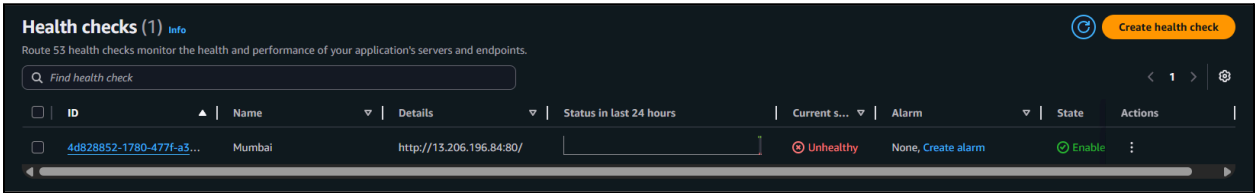
- Go to AWS Console → CloudFormation.
- Upload YAML template.
- Create CloudFormation stack.
- Verify infrastructure deployment.



Step 10: Test Disaster Recovery Failover

- Access the application through Route 53 domain routing.
- Verify that users are directed to the Mumbai primary server.
- Stop Apache service on Mumbai EC2 instance.
- Wait for Route 53 health check to detect failure.
- Verify traffic automatically redirects to Singapore DR server.

```
May 23 18:41:12 ip-172-31-47-198.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
May 23 18:41:12 ip-172-31-47-198.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
May 23 18:41:12 ip-172-31-47-198.ap-south-1.compute.internal httpd[28460]: Server configured, listening on: port 80
[root@ip-172-31-47-198 ec2-user]# sudo systemctl stop httpd
[root@ip-172-31-47-198 ec2-user]#
```



Troubleshooting:

Issue 1: Route 53 Health Check Shows Unhealthy

The health check may fail if the Apache web server is stopped or HTTP access is blocked by the security group.

Solution:

- Verify Apache service is running.
- Ensure HTTP port 80 is allowed in the EC2 security group.
- Check correct public IP configuration in Route 53 records.

Issue 2: Failover Routing Not Working

Failover may not function if primary and secondary records are configured incorrectly.

Solution:

- Ensure both records use the same record name.
- Attach health check only to the primary failover record.
- Verify Route 53 routing policy is configured as Failover.

Conclusion:

The Multi-Region Disaster Recovery Plan provides a reliable and scalable cloud architecture for maintaining application availability during failures or regional outages. By deploying resources across Mumbai and Singapore regions, the project demonstrates effective disaster recovery planning and automated failover capabilities. Amazon Route 53 health checks continuously monitor the primary server and automatically redirect users to the disaster recovery environment when required.

Using AWS CloudFormation, the infrastructure deployment process becomes automated and manageable. The project highlights important cloud computing concepts such as high availability, fault tolerance, failover automation, and multi-region architecture. This disaster recovery solution is suitable for enterprise applications that require continuous availability and minimal downtime.